---Question---

A video editor wants to preview footage smoothly at 120 FPS on a professional workstation. The system uses a 24-bit color depth per pixel for the frame buffer. A frame size in the buffer is 24,883,200 bytes. What should the GPU's minimum rendering speed (in pixels per **millisecond**) be to maintain this frame rate?

---Solution---

Color depth = 24 bit
Image size = 24883200*8 bits

120 Frames per second
So, 1 Frame takes 1/120 sec = 0.008*1000 ms = 8.33 ms

Image size = Resolution*Depth
-> Resolution = Image Size/Depth = 24883200*8/24 = 8294400 pixels

In 8.33ms 8294400 pixels are rendered
->In 1 ms = 8294400/8.33 = **995726.29 Pixels**